

Changes of White Matter Diffusivity After a Season of High School Varsity Football

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Introduction: American football is one of the most popular sports in the United States, despite having the highest concussion rate among contact sports [1]. Neurodegenerative diseases like Chronic Traumatic Encephalopathy have been linked to retired professional football athletes with extended repetitive head impact exposure (HIE) [2]. Diffusion tensor imaging (DTI) enabled the assessment of white matter (WM) microstructure, and is used to characterize brain injuries [1,3]. Evidence from sports-related concussions and nonconcussive impact studies have linked fractional anisotropy [FA] and mean diffusivity [MD] changes with various phases of mild traumatic brain injury (mTBI) [1,3]. However, the impact of repetitive HIE on the developing adolescent brain remains understudied despite the participation of over one million high school athletes in football across the United States [2]. To address this issue, we hypothesize that during one season of high school varsity football, WM microstructure will be affected even in the absence of clinically diagnosed concussions due to cumulative HIE. We will test this hypothesis by analyzing longitudinal diffusion MRI (FA, MD, axial diffusion [AD], and radial diffusivity [RD]) and cumulative HIE data.

Materials and Methods: The subjects were male high school athletes (age=15.8±1.3 years) with a mean time between assessments of 142.3±51.8 days. Biomechanical exposure data was acquired through the Head Impact Telemetry System (HITSTM), from which combined-probability risk-weighted exposure (RWE_{cp}) was computed [4]. This considers both rotational and linear accelerations experienced in head impacts. Subjects who had a concussion history 6 months prior and during the season were excluded. Fifty-two participants pre- and post-season imaging data sets were obtained on a 3-T MRI scanner. DTI sequences were collected using fifteen diffusion directions with b=1000/2000 each. Local FA, MD, the primary eigenvalue λ_1 which represents AD, and the single eigenvalues λ_2 and λ_3 , which represent RD were determined based on the Johns Hopkins University (JHU) WM Atlas [2]. Two-tail paired *t*-tests were performed on each of the forty-eight WM regions of interest (ROIs) to assess differences in FA, MD, AD and RD between pre- and post-season measurements. False Discovery Rate (FDR) correction was applied to each ROI measurement. For ROIs with statistically significant changes in FA and MD post- and pre-season ($p_{FDR} < 0.05$), linear regression models were fitted to examine their association with log-transformed RWE_{cp} while correcting for days between scans and age. As a secondary analysis on those ROIs, two-tail paired *t*-tests were computed using single eigenvalues λ_1 , λ_2 , and λ_3 to further assess the diffusion adaptations in the three main directions.

Results and Discussion:

Table 1. Two-tailed Paired t-test Results Comparing DTI Metrics Post- and Preseason.

ROI Name	FA		MD		AD		RD	
	t-statistic	p value	t-statistic	p value	t-statistic	p value	t-statistic	p value
Pontine Crossing Tract*	2.558	0.033	-3.461	0.004	-2.941	0.059	-3.364	0.005
Corticospinal Tract R	2.727	0.026	-2.625	0.023	-2.688	0.073	-2.462	0.032
Anterior Limb of Internal Capsule R*	2.546	0.033	-3.088	0.008	-0.769	0.668	-3.424	0.004
Anterior Limb of Internal Capsule L*	2.908	0.018	-3.563	0.004	-0.301	0.812	-4.198	0.001
Posterior Limb of Internal Capsule R*	2.376	0.043	-3.461	0.004	-1.555	0.288	-3.100	0.008
Posterior Limb of Internal Capsule L* ^w	2.353	0.043	-3.289	0.005	-2.029	0.143	-3.005	0.009
Retrolenticular Part of Internal Capsule R	2.649	0.030	-3.910	0.003	-2.974	0.059	-3.706	0.002
Retrolenticular Part of Internal Capsule L*	2.596	0.033	-3.401	0.004	-1.685	0.248	-3.404	0.004
Anterior Corona Radiata R	3.437	0.010	-3.482	0.004	-2.221	0.141	-3.751	0.002
Anterior Corona Radiata L	3.031	0.015	-4.415	0.002	-3.438	0.028	-4.163	0.001
Superior Corona Radiata R	2.931	0.018	-3.238	0.005	-2.127	0.141	-3.323	0.005
Superior Corona Radiata L	3.033	0.015	-4.358	0.002	-3.608	0.028	-4.011	0.001
Posterior Corona Radiata R*	3.368	0.010	-3.903	0.003	-2.052	0.143	-4.178	0.001
Posterior Corona Radiata L	3.473	0.010	-3.697	0.004	-2.194	0.141	-4.037	0.001
Posterior Thalamic Radiation R* ^w	3.805	0.009	-2.957	0.010	-0.283	0.812	-3.543	0.003
Sagittal Stratum R ^v	3.961	0.009	-3.744	0.004	-1.031	0.492	-4.379	0.001
External Capsule R*	2.397	0.042	-3.330	0.005	-2.397	0.121	-3.210	0.006
External Capsule L	2.455	0.038	-3.475	0.004	-2.807	0.068	-3.309	0.005
Cingulum Cingulate Gyrus L	2.320	0.045	-3.448	0.004	-0.648	0.708	-4.670	0.001
Superior Longitudinal Fasciculus R	3.692	0.009	-3.603	0.004	-1.732	0.238	-4.214	0.001
Superior Longitudinal Fasciculus L	3.121	0.014	-3.476	0.004	-2.067	0.143	-3.775	0.002

Note: p-values shown were FDR corrected.

*MD vs RWE_{cp}: $p_{FDR} < 0.05$; ^vFA vs RWE_{cp}: $p_{FDR} < 0.05$

As seen in Table 1, twenty-one out of the forty-eight WM ROIs demonstrated significant ($p_{FDR} < 0.05$) increases in FA and decreases in MD and RD. Only the left Anterior and Superior Corona Radiata showed significant decreases in AD. Linear regression models revealed more significant associations between delta MD and RWE_{cp} in comparison with FA (Table 1). Changes in MD were

negatively correlated to RWE_{cp}, conversely, changes in FA were positively correlated. Our linear regression models suggest that some DTI scalar changes may be related to the cumulative HIE players endure during the

season, while others may be due to normal physiological processes. The direction of DTI scalars in the ROIs that have significant correlations with RWEcp suggests axonal inflammation. This is denoted through the decrease in the average water diffusion, as evident in the reduction in MD and the increase of linearity (increases of FA) [3]. The paired t-tests results of the single eigenvalues suggested that FA and MD alterations are primarily due to decreases in λ_2 and λ_3 as most significant changes occurred in these eigenvalues (Figure 1).

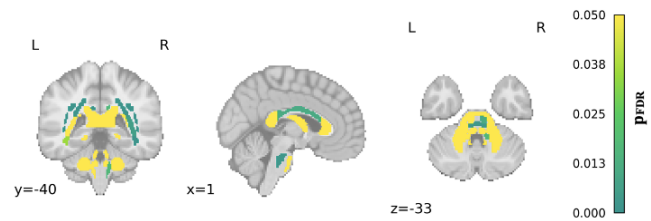


Figure 1. Two-tailed paired t-tests results of comparison of post- and preseason λ_2 and λ_3 .

Conclusions: This study demonstrated that, in a season of high school football, there are DTI metric changes: increases in FA and decreases in MD and RD. In some ROIs these MD differences had a significant relationship with cumulative HIE in the absence of clinically diagnosed concussions. To our knowledge, this is the first study of this population to report the directions of the diffusion changes examined via λ_1 , λ_2 , and λ_3 . Anisotropy in WM may have been expressed as there were significant λ_2 and λ_3 changes. This data adds to the growing evidence that, even in the absence of concussions during the season, there is measurable brain changes due to HIE. However, the results should be carefully interpreted due to limitations such as a small sample size and the absence of follow-up evaluations.

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